

Contents

1	Topological Space	2
1.1	Basic definition	2
1.2	Basis for a Topology	3
1.3	Closed Sets and Limit Points	7
1.3.1	Closure, Interior	7
1.3.2	Limit Points	8

Topological Space

1.1 Basic definition

Definition 1.1.1

Let X be a non-empty set. A set $\mathcal{T}$ of subsets of X is said to be a **topology** on X if

- (i) X and the empty set, $\emptyset$, belong to $\mathcal{T}$,
 - (ii) the union of any (finite or infinite) number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$, and
 - (iii) the intersection of any two sets in $\mathcal{T}$ belongs to $\mathcal{T}$.
- The pair $(X, \mathcal{T})$ is called a **topological space**.

Definition 1.1.2

Let $(X, \mathcal{T})$ be any topological space. Then the members of $\mathcal{T}$ are said to be **open sets**.

Proposition 1.1.3

If $(X, \mathcal{T})$ is any topological space, then

- (i) X and $\emptyset$ are open sets,
- (ii) the union of any (finite or infinite) number of open sets is an open set,
- (iii) the intersection of any finite number of open sets is an open set.

Definition 1.1.4

Let $(X, \mathcal{T})$ be a topological space. A subset S of X is said to be a **closed set** in $(X, \mathcal{T})$ if its complement in X , namely $X \setminus S$, is open in $(X, \mathcal{T})$.

Proposition 1.1.5

If $(X, \mathcal{T})$ is any topological space, then

- (i) $\emptyset$ and X are closed sets,
- (ii) the intersection of any (finite or infinite) number of closed sets is a

closed set and
 (iii) the union of any finite number of closed sets is a closed set.

Definition 1.1.6

A subset $\mathcal{S}$ of a topological space $(X, \mathcal{T})$ is said to be **clopen** if it is both open and closed in $(X, \mathcal{T})$.

1.2 Basis for a Topology

Definition 1.2.1

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of open subsets of X is said to be a **basis** for the topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

Remark.

拓扑基不是唯一的, 根据定义, 一旦一个集族 $\mathcal{B}$ 成为了一个拓扑基, 我们无论往 $\mathcal{B}$ 添加多少开集, 它还是这个拓扑的基. 因此直觉上, 拓扑的基是越小越好用的.

Note.

也就是说, 拓扑基是通过任意并运算生成拓扑的对象.

Proposition 1.2.2

Let X be a non-empty set and let $\mathcal{B}$ be a collection of subsets of X . Then $\mathcal{B}$ is a basis for a topology on X if and only if $\mathcal{B}$ has the following properties:

- (a) $X = \bigcup_{B \in \mathcal{B}} B$, and
- (b) for any $B_1, B_2 \in \mathcal{B}$, the set $B_1 \cap B_2$ is a union of members of $\mathcal{B}$.

Remark.

这里不谈与 X 上的任何一个具体的拓扑的关系, 只谈一个集族什么时候能够生成一个拓扑.

Note.

这个命题刻画了一个集族距离成为一组基有多远. 即一个集族需要什么额外条件, 才能通过任意并这个运算生成出一个拓扑. 命题中的条件 (a) 无非对应于 Definition 1.1.1 (i). 而通过基任意并生成的这个操作, 天然就是为了 Definition 1.1.1 (ii) 而建立的. 因此 (b) 就是为了满足 1.1.1 (iii) 而提出的条件. 我们发现生成拓扑对于有限交封闭的要求, 完全地变成了对于基的有限交的封闭的要求.

Proof. If $\mathcal{B}$ is a basis for a topology on X , say $\mathcal{T}_{\mathcal{B}}$. Then $X \in \mathcal{T}$ means that there is a collection of sets in $\mathcal{B}$, say $\mathcal{B}_1$, such that

$$X = \bigcup_{B \in \mathcal{B}_1} B \subseteq \bigcup_{B \in \mathcal{B}} B$$

And since every sets in $\mathcal{B}$ is a subset of X , it follows that (a) holds. To show (b), we only need to see that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Since $\mathcal{T}$ is closed under finite intersection, we have $B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$.

Then conversely, if $\mathcal{B}$ has the properties (a), (b), we only show that the collection $\langle \mathcal{B} \rangle$ of union of the members in $\mathcal{B}$, satisfies Definition 1.1.1 (iii). We take $A_1, A_2 \in \langle \mathcal{B} \rangle$, and suppose that

$$A_i = \bigcup_{j \in I_i} B_j, \quad B_j \in \mathcal{B}, \quad I_i \text{ is a index set, } i = 1, 2$$

Then

$$\begin{aligned} A_1 \cap A_2 &= \left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap \left(\bigcup_{j_2 \in I_2} B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left(\left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left(\bigcup_{j_1 \in I_1} (B_{j_1} \cap B_{j_2}) \right) \\ &= \bigcup_{j_1 \in I_1, j_2 \in I_2} B_{j_1} \cap B_{j_2} \end{aligned}$$

Since every $B_{j_1} \cap B_{j_2}$ is a union of members of $\mathcal{B}$, and $A_1 \cap A_2$ is a union of such $B_{j_1} \cap B_{j_2}$, then $A_1 \cap A_2$ is a union of members of $\mathcal{B}$. $\square$

Proposition 1.2.3

Let $(X, \mathcal{T})$ be a topological space. A family $\mathcal{B}$ of open subsets of X is a basis for $\mathcal{T}$ if and only if for any point x belonging to any open set U , there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark.

表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “ U 可以由 $\mathcal{B}$ 通过 ‘并’ 生成” 的微观讲法. 带入这句话之下, 这个命题几乎就是一句 “废话”. 同样的道理也出现在 Proposition 1.2.4

Note.

想要看 $\mathcal{B}$ 是否作为基生成了拓扑 $\mathcal{T}$. 我们不用谈 $\mathcal{B}$ 中的元素如何如何 “并” 出每一个开集 U . 而是只需要 $\mathcal{B}$ 对于开集是足够 “精细” 的, 以至于对于任意的开集 U , 我们可以用 $\mathcal{B}$ 里面的元素在 U 上 “框出” 它的任意一个点.

Proof. If $\mathcal{B}$ is a basis for $\mathcal{T}$. Then we can write U as

$$U = \bigcup_{i \in I} B_i$$

Then for any $x \in U$, there exists $i_0 \in I$ such that $x \in B_{i_0} \subseteq U$.

Now we show the converse. First we show that $\mathcal{B}$ is a basis. We use Proposition 1.2.2 to verify this. Since X is a open set, for any $x \in X$, there is a $B_x \in \mathcal{B}$, such that $x \in B_x \subseteq X$, we get

$$X = \bigcup_{x \in X} B_x \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq X$$

Thus

$$X = \bigcup_{B \in \mathcal{B}} B$$

Take any $B_1, B_2 \in \mathcal{B}$, we know $B_1 \cap B_2$ is again an open set. Then similarly, we get

$$B_1 \cap B_2 = \bigcup_{x \in B_1 \cap B_2} B_x$$

is a union of members in $\mathcal{B}$. Thus $\mathcal{B}$ is actually a basis. Finally, we only need to show that $\mathcal{B}$ generates $\mathcal{T}$, that is $\langle \mathcal{B} \rangle = \mathcal{T}$. $\langle \mathcal{B} \rangle$ is the collection of unions

of members in $\mathcal{B}$. $\langle \mathcal{B} \rangle \subseteq \mathcal{T}$ is obvious. $\mathcal{T} \subseteq \langle \mathcal{B} \rangle$ concludes from

$$U = \bigcup_{x \in U} B_x \in \langle \mathcal{B} \rangle$$

□

Proposition 1.2.4

Let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on a set X . Then a subset U of X is open if and only if for each $x \in U$ there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Note.

把基 $\mathcal{B}$ 看成是 $\mathcal{T}$ 中代表着“足够精细”的某一批. 验证 U 是否是开集, 只需要看 U 是否可以被这一小批集合“概括”.

Proof. $\implies$ is immediately from Proposition 1.2.3. Now suppose that for each $x \in U$, there exists a $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq U$, then

$$U = \bigcup_{x \in U} B_x \in \langle \mathcal{B} \rangle = \mathcal{T}$$

□

Proposition 1.2.5

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a non-empty set X . Then $\mathcal{T}_1 = \mathcal{T}_2$ if and only if

1. for each $B \in \mathcal{B}_1$ and each $x \in B$, there exists a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$, and
2. for each $B \in \mathcal{B}_2$ and each $x \in B$, there exists a $B' \in \mathcal{B}_1$ such that $x \in B' \subseteq B$.

Proof sketch.

考虑这句话: 表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “ U 可以由 $\mathcal{B}$ 通过 ‘并’ 生成”. 1. 是说 $\mathcal{B}_1$ 中任意集合可以由 $\mathcal{B}_2$ “并” 出来. 自然地 $\mathcal{T}_1$ 中任意集合也被 $\mathcal{B}_2$ “并出来”. 因此 1. 就是 $\mathcal{T}_1 \subseteq \mathcal{T}_2$. 同样地, 2. 就是 $\mathcal{T}_2 \subseteq \mathcal{T}_1$.

Lemma 1.2.6

Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively, on X . Then the following are equivalent:

1. $\mathcal{T}'$ is finer than $\mathcal{T}$.
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x , there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Proof sketch.

同样的, 2. 就是在说每个 B 可以由一堆 B' 并出来. 那么显然每个 $U \in \mathcal{T}$ 可以由一堆 B' 并出来, 即 $U' \in \mathcal{T}'$.

1.3 Closed Sets and Limit Points

Definition 1.3.1

A subset A of a topological space X is said to be **closed** if the set $X - A$ is open.

1.3.1 Closure, Interior

Definition 1.3.2

Given a subset A of a topological space X

- The **interior** of A , denoted by A° or by $\text{Int } A$, is defined as the union of all open sets contained in A .
- The **closure** of A , denoted by $\bar{A}$ or by $\text{Cl } A$, is defined as the intersection of all closed sets containing A .

Theorem 1.3.3

Let A be a subset of the topological space X .

- (a) Then $x \in \bar{A}$ if and only if every open set U containing x intersects A .
- (b) Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A .

Proof. (a) Method 1. Note that

$$V \text{ is a closed set containing } A \iff V^c \text{ is an open set such that } V^c \cap A = \emptyset$$

We have

$$\begin{aligned}
 x \notin \bar{A} &\iff x \notin \bigcap_{V \text{ is a closed set containing } A} V \\
 &\iff x \in \bigcup_{V \text{ is a closed set containing } A} V^c \\
 &\iff x \in \bigcup_{U \text{ is an open set such that } U \cap A = \emptyset} U \\
 &\iff \text{there exists open set } U \text{ such that } x \in U, U \cap A = \emptyset
 \end{aligned}$$

Method 2. Suppose that $x \in \bar{A}$, then x is contained in any closed set containing A . If there exists open set U containing x does not intersect A . Then U^c is a closed set containing A , which implies that $x \in U^c$, a contradiction. Conversely, if every open set containing x intersects A , then every closed set that x does not lie in cannot contain A . i.e. x lies in every closed set containing A .

- (b) If every open set containing x intersects A , then every basis B containing x intersects A since B is also an open set. Conversely, if every basis B containing x intersects A , since every open set U contains a basis containing x , it intersects with A as well. Then it follows from (a).

□

1.3.2 Limit Points

Definition 1.3.4

If A is a subset of the topological space X and if x is a point of X , we say that x is a **limit point** (or “cluster point,” or “point of accumulation”) of A if every neighborhood of x intersects A in some point **other than x itself**. Said differently, x is a limit point of A if it belongs to the closure of $A - \{x\}$.

Theorem 1.3.5

Let A be a subset of the topological space X ; let A' be the set of all **limit points** of A . Then

$$\bar{A} = A \cup A'.$$

Proof sketch.

对于 $\bar{A} \subseteq A \cup A'$, $A \subseteq \bar{A}$ 显然而 $A' \subseteq A$ 是 Theorem 1.3.3 的直接推论. $A \cup A' \subseteq \bar{A}$, 则考虑若 x 不在 A 里面, 那么 $x \in A'$ 的定义中的开集一定与 A 相交, 从而有 $x \in \bar{A}$.